

The Unit Circle: Linking Angles, Coordinates and Trig Ratios

SimuPhysics Unit Circle Simulation · SimuPhysics

Senior Secondary (Years 11-12)	~30 minutes	Mathematics · Trigonometry
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Simulation: <https://simuphysics.com/view/mathematics-unit-circle-simulation.html>
Curriculum links: trigonometry

Learning intentions

- I can locate an angle on the unit circle and read off its coordinates.
- I can explain why the x- and y-coordinates of a point on the unit circle equal cosine and sine of the angle.
- I can predict the sign of sine and cosine in each quadrant of the unit circle.

Getting started

Open the SimuPhysics Unit Circle Simulation. Drag the point around the circle to 0 degrees, and check that its coordinates read (1, 0). Keep the angle, coordinate and trig-ratio displays visible as you work.

Tasks

1. Drag the point to 90 degrees. Record its coordinates and compare them with $\sin(90^\circ)$ and $\cos(90^\circ)$.

2. Drag the point to 0° , 30° , 90° , 180° and 270° , and record its coordinates alongside sin and cos of each angle.

Angle	x-coordinate	y-coordinate	$\sin(\text{angle})$	$\cos(\text{angle})$

3. Drag the point into the second quadrant, between 90° and 180° . What happens to the sign of the x-coordinate? What does this tell you about cosine in this quadrant?

4. Using your table, explain the relationship between a point's coordinates on the unit circle and the sine and cosine of its angle.

5. Without dragging the point, predict the coordinates at 150° using symmetry with the 30° position. Then check your prediction against the simulation.

Working:

6. Explain why sine and cosine values are always between -1 and 1, using the unit circle as your justification.

7. Drag the point to an angle where sine and cosine are equal in value. What angle is this, and why does this make sense geometrically?

Extension (optional)

Explain how the unit circle could be used to define sine and cosine for angles greater than 360° degrees, or negative angles, which don't correspond to an angle in a standard right-angled triangle.
